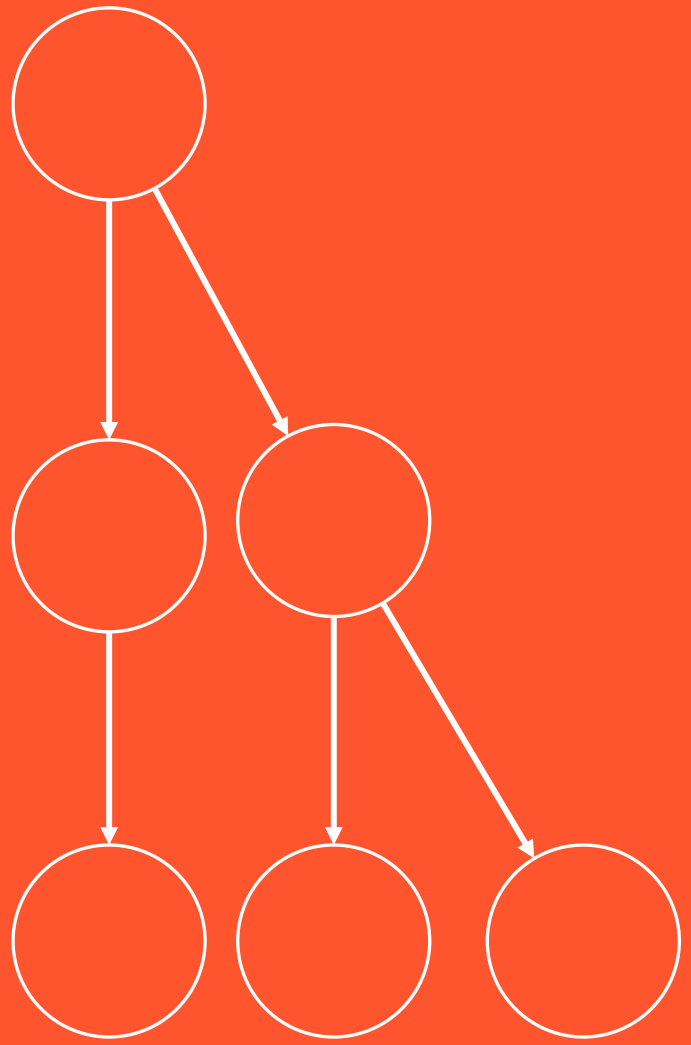


BCOG 100

Vertebrate Neural Architecture

Lecture 6.3 — The Vertebrate Brain

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: Why Release Rather Than Switch On?

In your group, three minutes.

In the last lecture, choosing an action turned out to be *releasing* it, not switching it on.

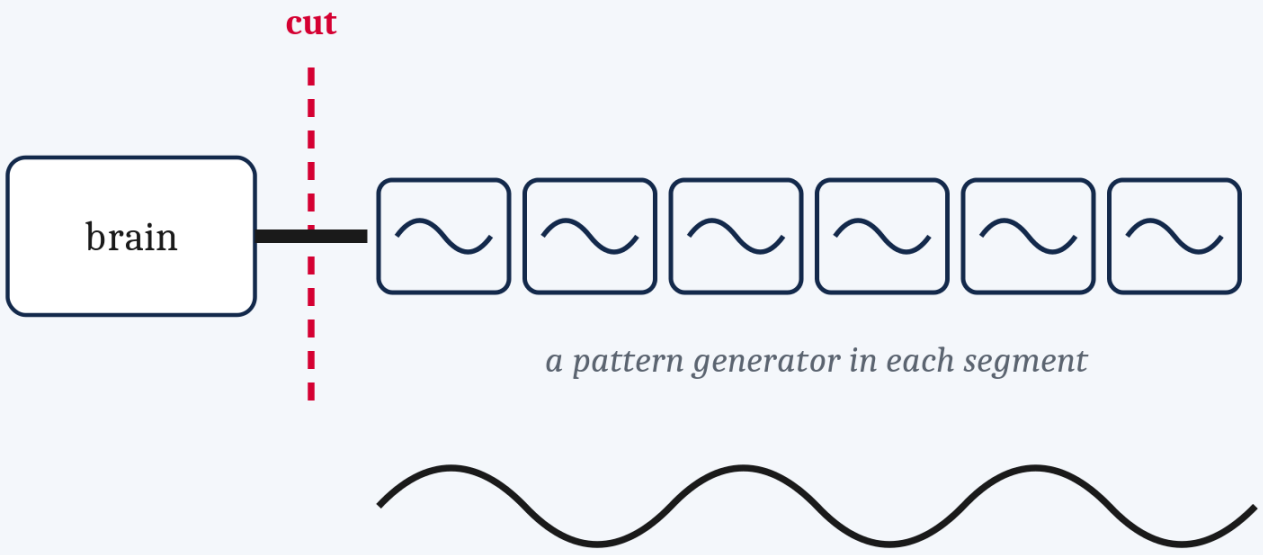
What does that mean, in a sentence?

And why is there no walking organ, no eating organ, no talking organ?

Hindbrain and Spinal Cord: Running the Body

The oldest-looking parts do the jobs that cannot stop. A **central pattern generator** makes a rhythm with no rhythmic input; the **reticular formation** sets how awake an animal is.

A lamprey's cord, cut from its brain



the swimming rhythm is still produced

What the hindbrain keeps running

- **breathing, heart rate,** blood pressure, swallowing
- **the reticular formation:** how awake, asleep or alert
- **one small cluster:** most of the brain's norepinephrine

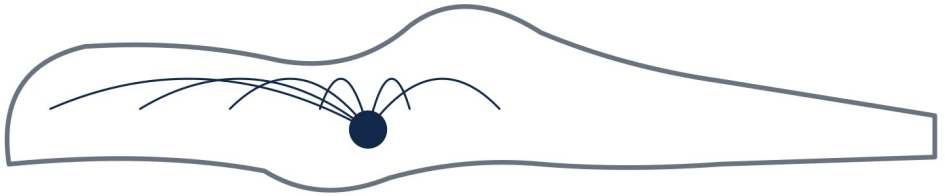
Down here, dedicated circuitry for a single behavior really does exist.

The Midbrain: Orienting, and Where the Broadcasts Start

The tectum aims by finding a place on a map. And the broad projections of the last lecture start in a few small clusters.

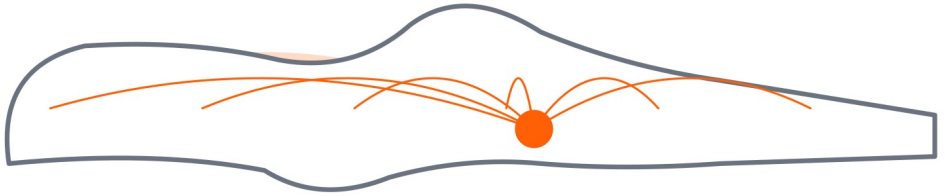
Three small places, and how far their fibers reach

dopamine
a cluster in the midbrain



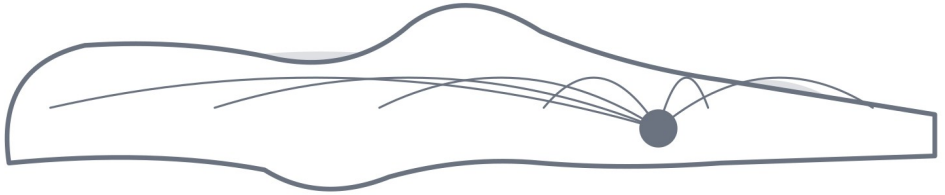
reaches
the forebrain, mostly

serotonin
cells along the midline



reaches
most of the brain

norepinephrine
one cluster, in the hindbrain

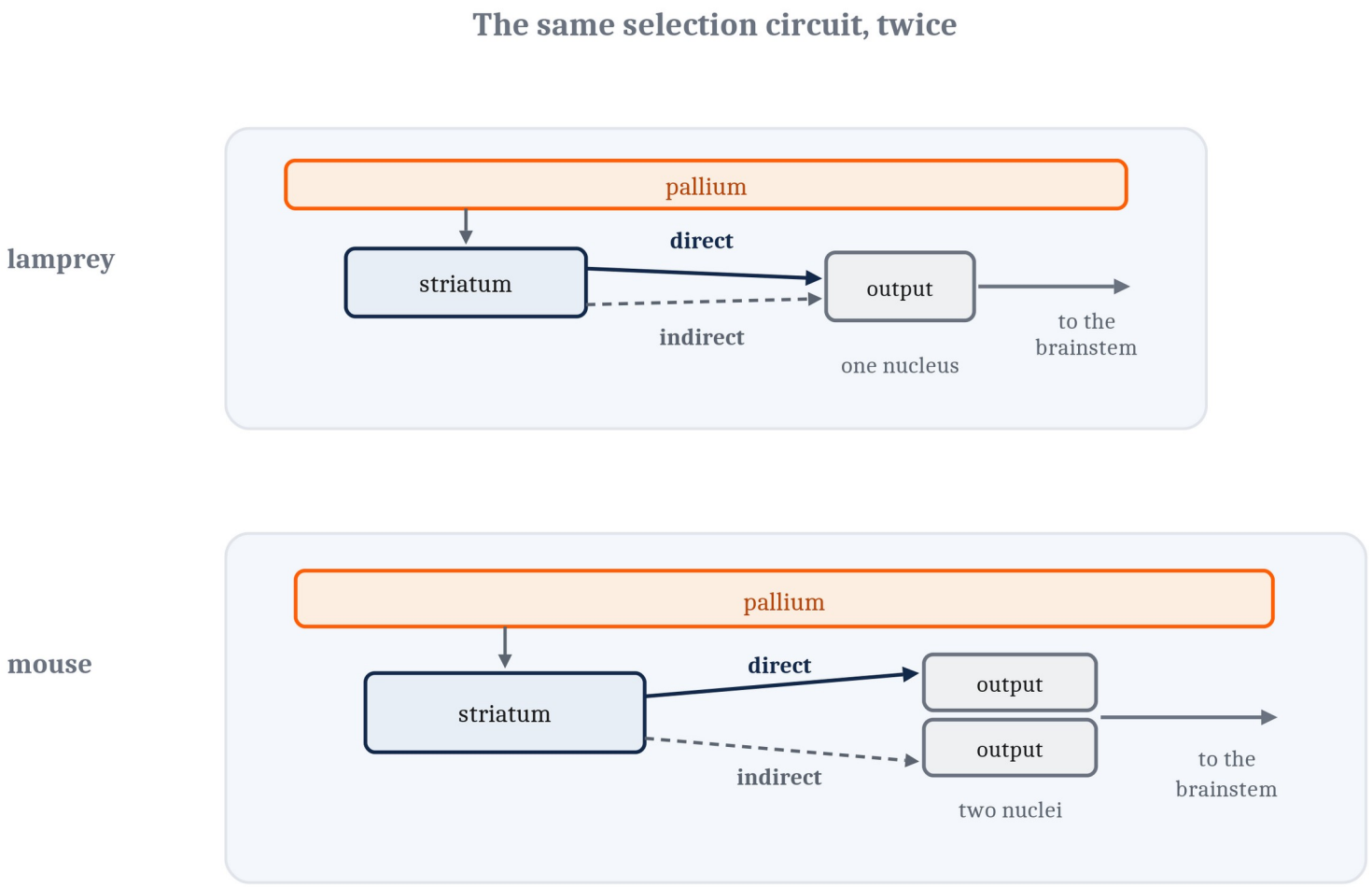


reaches
most of the brain

the same brain three times; the shading is the territory each set of fibers reaches

The Forebrain's Old Core

Homeostasis from the hypothalamus, selection from the basal ganglia, and valence from the **amygdala**, the last in jawed vertebrates.

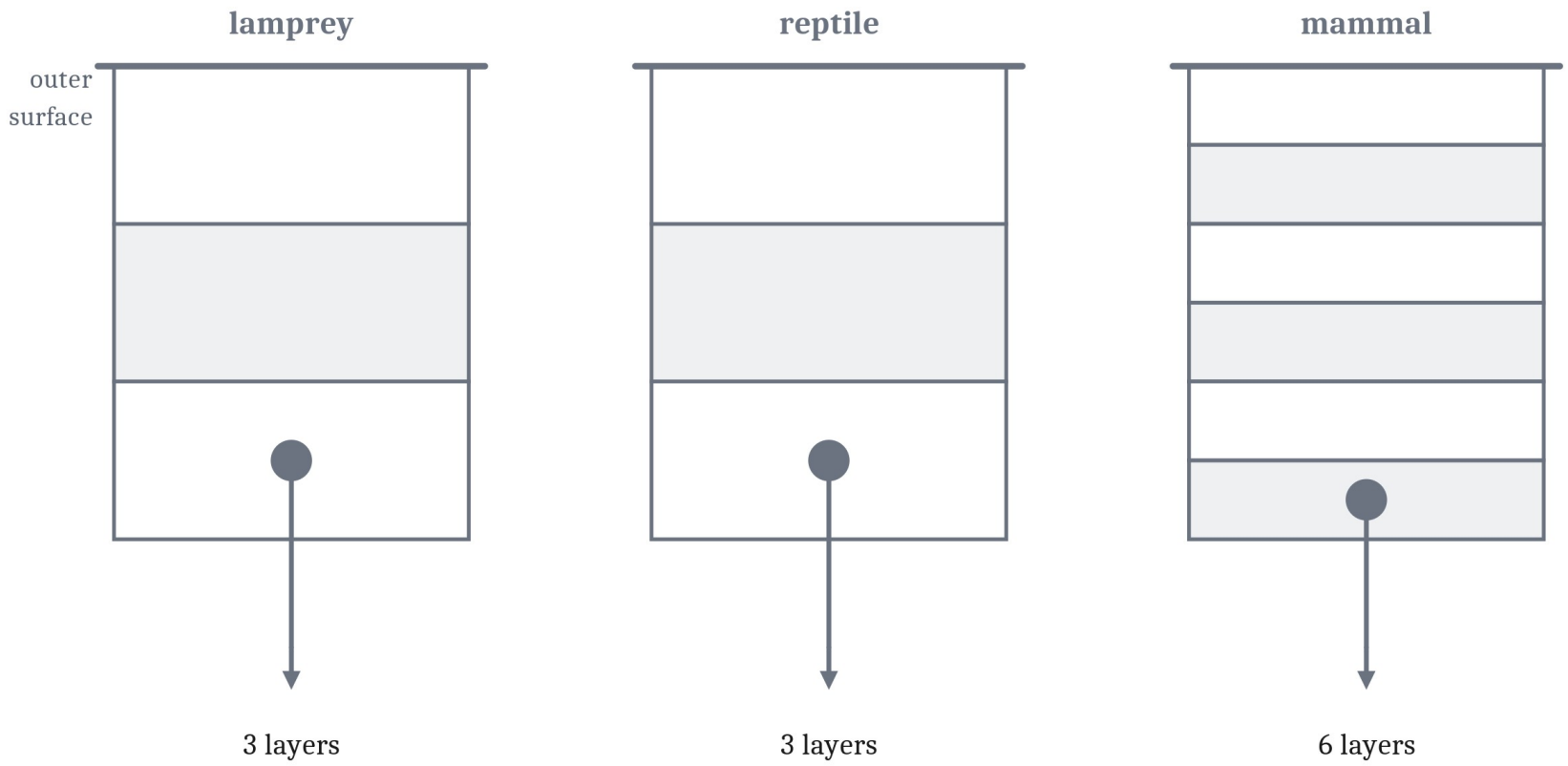


same transmitters, same connections, same two routes — what differs is how many copies

The Pallium, and the Thalamus That Serves It

Every vertebrate has a pallium, built differently in each. Almost everything that reaches it passes through the thalamus.

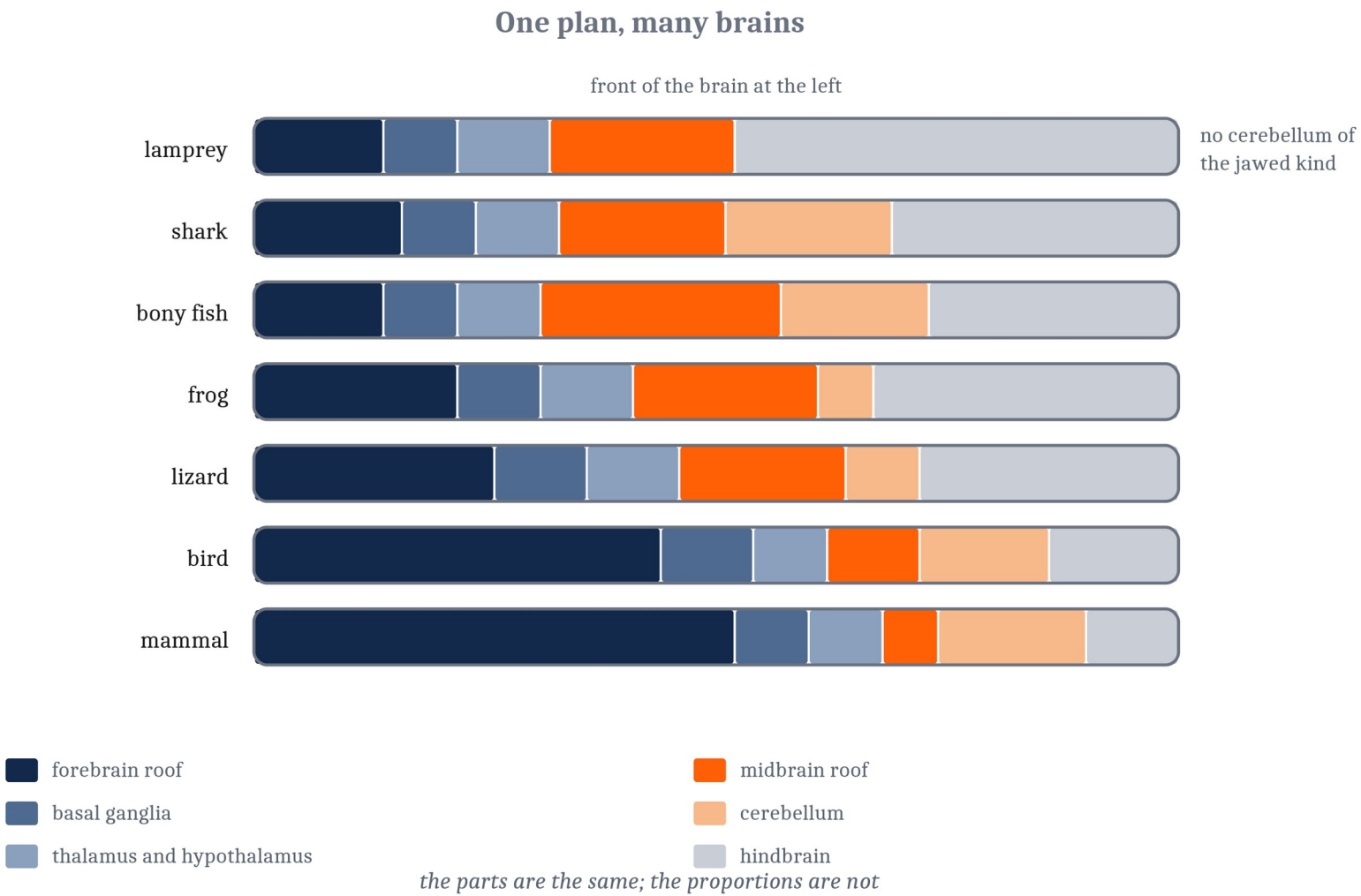
Three layers, three layers, six



the cell that sends commands down to the brainstem and cord is in all three

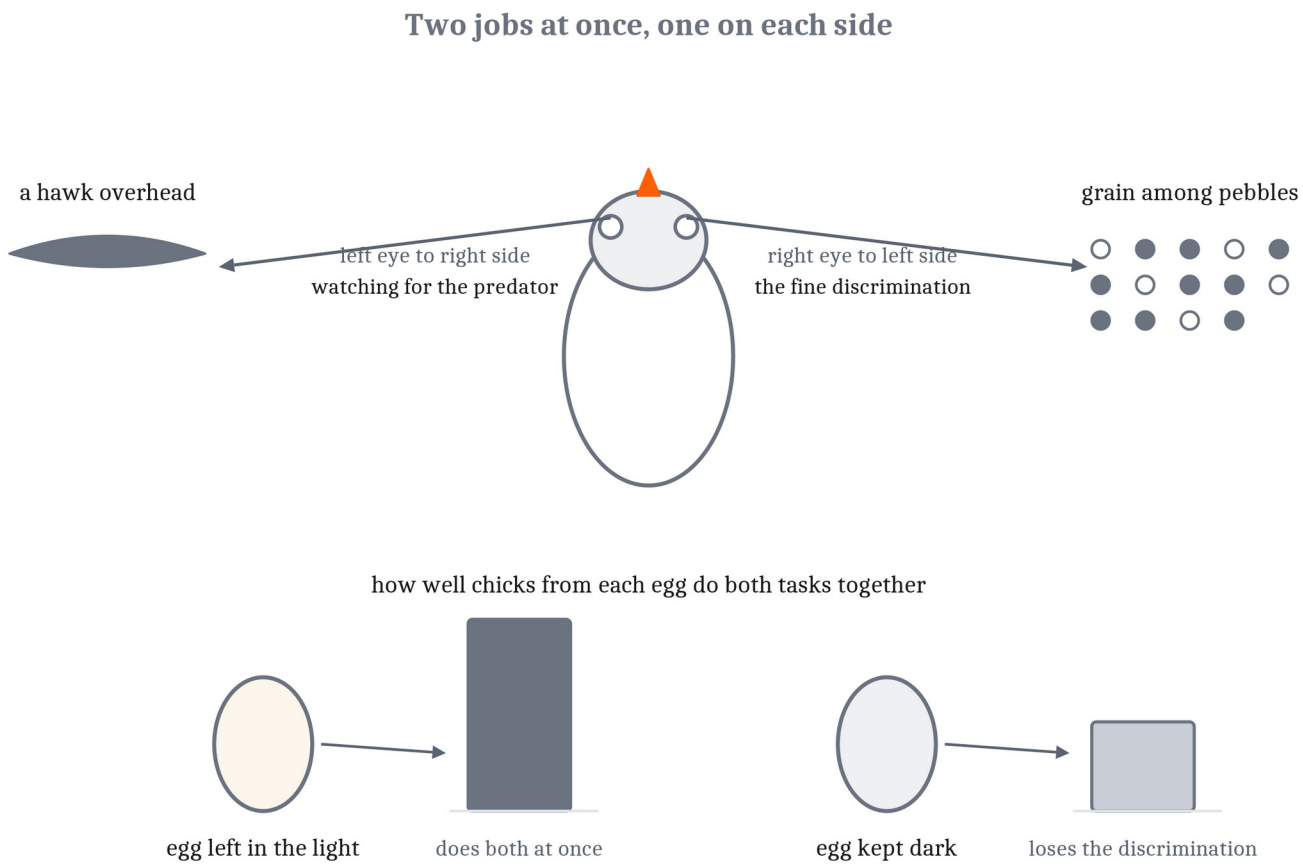
One Plan, Elaborated Many Times Over

The same parts, in the same order, from lamprey to mammal. What changed is proportion and number, and it changed **in parallel, not in one climb**.



Two Halves

Lateralization: the two sides of a brain doing different things. The chick shows why an individual benefits. Why a whole population leans the same way is a separate question.

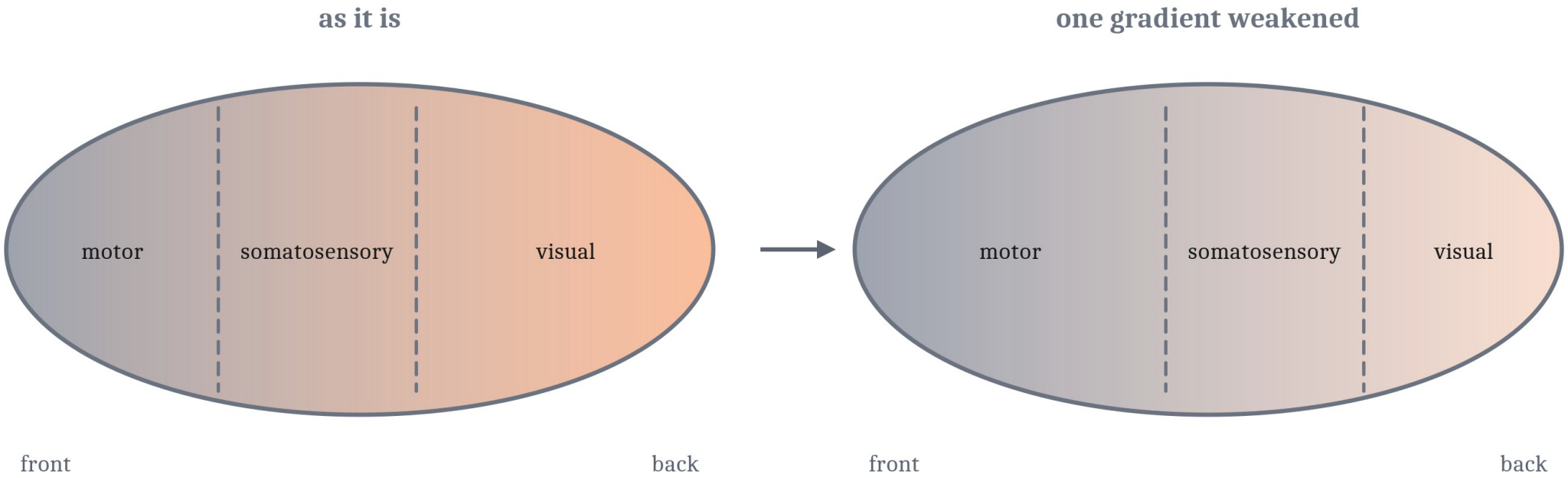


light through the shell reaches the right eye only, and strengthens the asymmetry

Rules, Not a Blueprint

No genome lists which cell goes where. It holds a few rules run locally: read a gradient, listen to the neighbors, follow a cue, compete for a limited supply.

Two gradients, and the areas that fall out of them



where a cell sits in the pair of gradients sets which area it belongs to

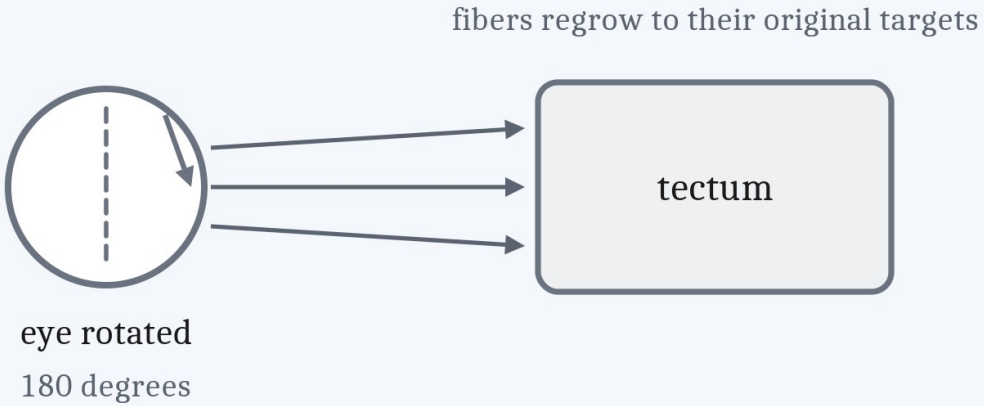
shift one gradient and every boundary moves together

Made, Moved, Steered, and Culled

A neuron's birthday sets its layer. An axon finds its target by a position, not an address. Too many neurons are made, and the target settles which survive.

Sperry's frog, and what was found instead of labels

the experiment

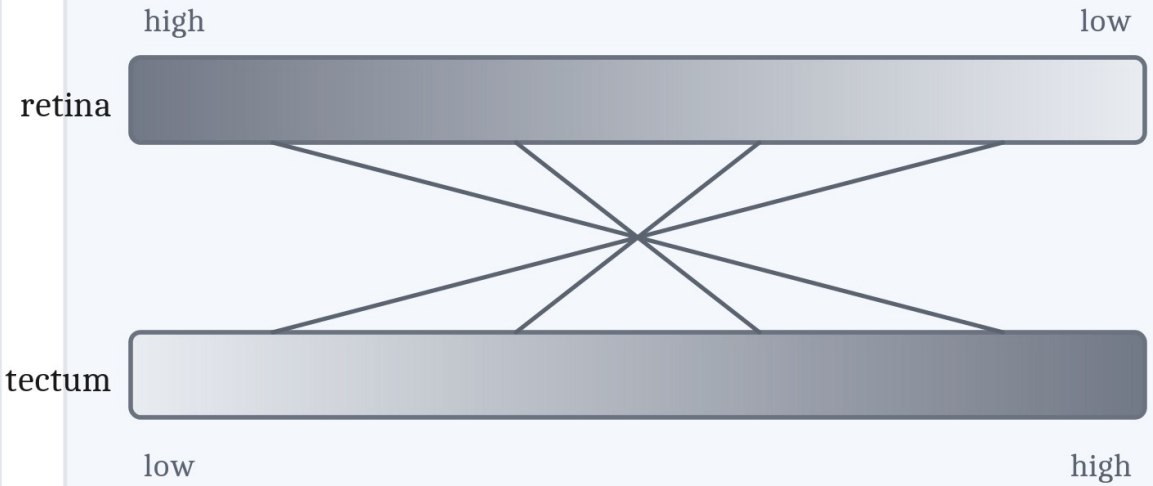


so the frog strikes to the wrong side

a unique label per cell would explain that

and so would something much cheaper

what was found

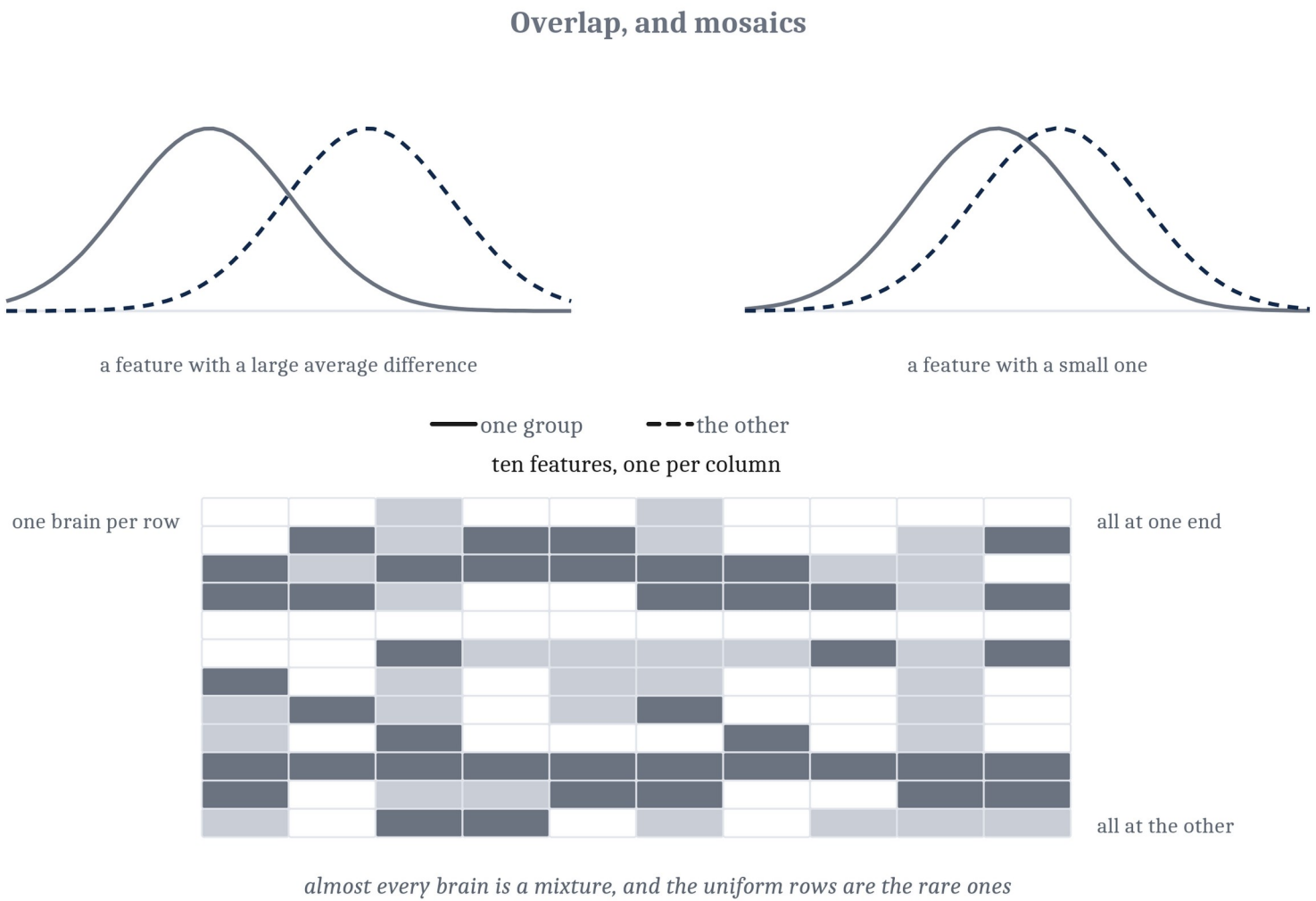


each fiber ends where its level in one gradient matches the level in the other

a position, not an address

Sex Differences as Development

Each region responds to signals in its own way, so the mechanism predicts a **mosaic**, not two kinds of brain. Across more than fourteen hundred human brains, that is what was found.



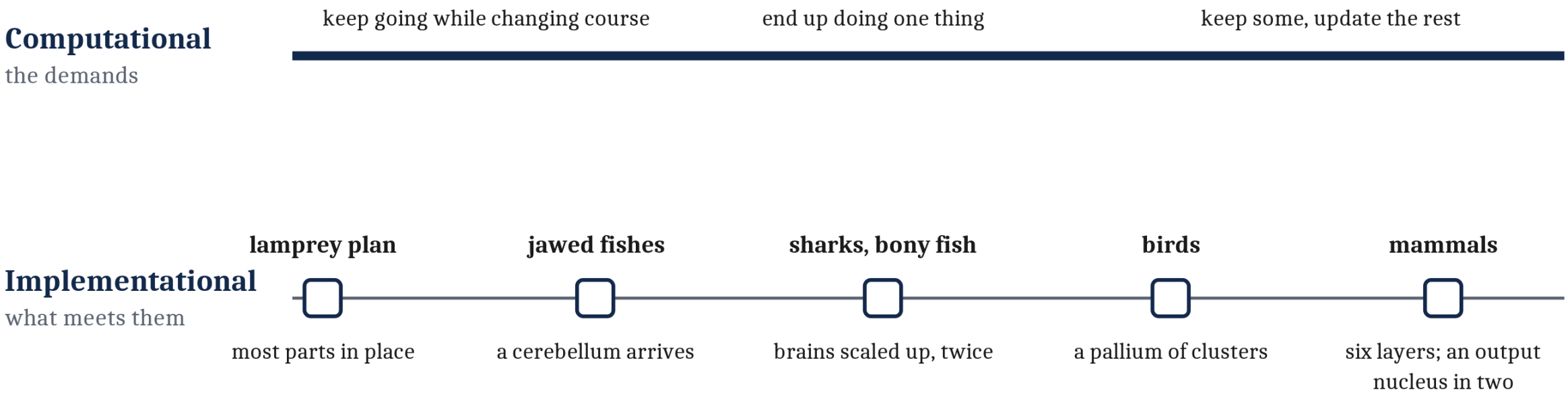
An Address Is Not an Account

Structures and functions map **many to many**. Neither list below is a list of separate functions done by separate parts.

Structure	Jobs it has been credited with
Amygdala	Learning that something predicts a threat; learning that something predicts a reward; responding to whatever is salient, good or bad; adjusting attention; influencing what is remembered
Striatum	Releasing one action and holding back others; learning which actions lead to reward; habits, as behavior becomes automatic; keeping track of how well things are going

Old Problems, New Parts

The computational level barely moved. The implementational level was rebuilt again and again. So finding an arrangement does not tell us which demand it answers.



elaborated many times over, in parallel, not climbed once toward mammals

A problem can stand still while the machinery that answers it is remade several times over.

Headlines About the Brain's "Centers"

In your group, four minutes.

Headlines regularly announce that a feeling, a trait or a habit has been found in one part of the brain.

Invent or recall one. What inference is it making?

What would a study have to show to earn the word *center*?

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3	What cannot stop	fig-cpg.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
4	Three small places, and how far their fibers reach	fig-brainstem-sources.png	CC BY 4.0	Jon Willits — drawn for the Module 6 reading · own work
5	The same selection circuit, twice	fig-lamprey-vs-mouse-bg.png	CC BY 4.0	Jon Willits — drawn for the Module 6 reading · own work
6	Three layers, three layers, six	fig-pallium-layers.png	CC BY 4.0	Jon Willits — drawn for the Module 6 reading · own work
7	One plan, many brains	fig-one-plan-many-brains.png	CC BY 4.0	Jon Willits — drawn for the Module 6 reading · own work
8	Two jobs at once, one on each side	fig-chick-dual-task.png	CC BY 4.0	Jon Willits — drawn for the Module 6 reading · own work
9	Two gradients, and the areas that fall out of them	fig-arealization.png	CC BY 4.0	Jon Willits — drawn for the Module 6 reading · own work
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11	Overlap, and mosaics	fig-mosaic.png	CC BY 4.0	Jon Willits — drawn for the Module 6 reading · own work
13	One level stood still while another was rebuilt	fig-levels-came-apart.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work